

Evaluating Limits

Date_____ Period_____

Evaluate each limit.

1) $\lim_{x \rightarrow 0} \frac{1 - \sin\left(\frac{\pi}{2} - x\right)}{x}$

0

2) $\lim_{x \rightarrow 0} \frac{\cos\left(\frac{\pi}{2} - x\right)}{x}$

1

3) $\lim_{x \rightarrow 0} \frac{\tan(x)}{3x}$

 $\frac{1}{3}$

4) $\lim_{x \rightarrow 0} \frac{\sin(x)}{\sin(4x)}$

 $\frac{1}{4}$

5) $\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{4x}$

0

6) $\lim_{x \rightarrow 0} \frac{\sin^2(2x)}{x^2}$

4